UNIT 2 · REGRESSION ANALYSIS

Normal Linear Models

Maximum Likelihood Estimation, Coefficient Interpretation, and Inference in the Linear Model

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The normal model: a special case of GLM

GLM is a model where Y belongs to the exponential family.
When Y is normal, that particular case is what we call a **Normal Linear Model**

Generalized Linear Model

$$g(\mu_i) = x_i^T \beta$$

Y_i belongs to the exponential family

$g(.)$ is the link function



link = identity

Normal Linear Model

$$E(Y_i) = \mu_i = x_i^T \beta \;\; ; \;\; Y_i \sim N(\mu, \sigma^2)$$

$$y = X\beta + e \, , \;\; e_i \sim N(0, \sigma^2) \text{ i.i.d.}$$

The link function $g(\mu) = \mu$ doesn't transform anything: that's why the model looks "linear and direct," unlike logistic or Poisson.

Normal Linear Models

$$y = X\beta + e$$
$$Y \sim \text{Normal}$$

Multiple Regression

Primary objective: explain or predict a continuous outcome.

Predictors: can be continuous, categorical, or both.

Example: Predict monthly sales using advertising spend and store size.

ANOVA

Primary objective: compare the mean of a continuous outcome across groups.

Predictors: categorical variables only.

Example: Do average monthly sales differ across store formats?

ANCOVA

Primary objective: compare group means while adjusting for a continuous covariate.

Predictors: categorical variables + continuous covariates.

Example: Do average monthly sales differ across store formats after adjusting for advertising spend?

ANOVA

A BIT OF HISTORY

It all started with Fisher

Ronald A. Fisher
(1890–1962)

WHERE ANOVA COMES FROM

A real agricultural problem

In the 1920s, Fisher worked as a statistician at the Rothamsted Experimental Station in England, analyzing decades of crop yield data from field trials.

The problem

Agronomists tested many treatments at once (fertilizers, seed varieties, irrigation) and needed to know **which ones truly affected** crop yield.

The old approach was flawed

Comparing **every pair of groups with separate t-tests** inflates the chance of a **false positive** as more groups are added.

The solution: 1925

In "Statistical Methods for Research Workers", Fisher introduced **analysis of variance**: one single test to **compare all groups at once.**

The F statistic is named "F" in Fisher's honor, by statistician George Snedecor.

THE CONCEPT

What is ANOVA?

ANalysis Of VAriance

DEFINITION

A statistical technique used to **compare the means of two or more groups at the same time**, to determine whether the **differences observed between them are statistically significant** or simply due to random chance.

In other words, it answers one central question:

"Is at least one group different from the rest or are they all, in practice, the same?"



Compares 2+ groups at once

A single test instead of many pairwise comparisons



Real effect vs. random noise

Separates a genuine difference from ordinary sample variation



Extends the t-test

The t-test only compares 2 groups; ANOVA generalizes to any number k

DEFINING THE PROBLEM

What exactly does ANOVA compare?

Y **Response variable (continuous / quantitative)**

Example: the weight of the plants

X **Factor(s) (categorical)**

Example: treatments applied (fertilizers, seed varieties, irrigation, none)

Weight (Kg)

Observation	Treatment			
	None	Fertilizer	Seed Varieties	Irrigation
1	0.35	0.24	0.23	2.03
2	0.22	0.16	0.14	1.27
3	0.16	0.07	0.34	1.02
4	0.37	0.18	0.16	1.17
5	0.22	0.08	0.15	2.3

FACTOR

The number of independent (categorical) variables being analyzed

LEVEL

The number of categories or treatments within a factor

ANOVA 1 Factor 4 Levels

- **ANOVA Fixed Effects**

Define the observations and treatments to consider in the analysis

- **ANOVA Random Effects**

Consider all the treatments and random observations in the analysis

ASSUMPTIONS

- Dependent variable is quantitative
- Independency
- Randomness
- Normality
- Homoscedasticity

REAL-WORLD APPLICATIONS

Where is ANOVA used?

Any time we need to compare a continuous outcome across categorical groups:

Marketing / Advertising

Y: first-purchase order value (MXN)

X: acquisition channel (Instagram, Google, TikTok, Email)

ANALYSIS QUESTION

Does the acquisition channel affect how much a new customer spends on their first purchase?

Human Resources

Y: units produced / hour

X: work shift (AM, PM, night)

ANALYSIS QUESTION

Does the work shift affect employee productivity?

Manufacturing / Quality

Y: tensile strength (kg/cm²)

X: raw material supplier

ANALYSIS QUESTION

Does the supplier affect the quality of the raw material received?

Banking / Finance

Y: credit line amount (MXN)

X: employment type

ANALYSIS QUESTION

Does employment type affect the credit line a customer is assigned?

Healthcare

Y: recovery time (days)

X: treatment (A, B, placebo)

ANALYSIS QUESTION

Does the treatment received affect a patient's recovery time?

Customer Retention

Y: churn / degradation rate (%)

X: customer generation

ANALYSIS QUESTION

Does the customer's generation affect their likelihood of churning?

Common pattern: continuous Y + categorical X + "are the group means different?"

Every observation has an address

Weight (Kg)

Observation	Treatment			
	None	Fertilizer	Seed Varieties	Irrigation
1	0.35	0.24	0.23	2.03
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$\boldsymbol{T_i}$= Each treatments

$\boldsymbol{k}$ = Number of total treatments

$\boldsymbol{n_i}$= Number of observations in each treatment

$\boldsymbol{N}$= Number of total observations

$\boldsymbol{y_{ij}}$= Observation j in the i treatment

$\boldsymbol{y_i}$ = Sum of observations in the i treatment

$\boldsymbol{Y}$= Sum of all observations

$\boldsymbol{\bar{y}_i}$ = Average of the observations in the i treatment

$\boldsymbol{\bar{Y}}$ = Average of the $\bar{y}_i$

T_1	T_2	…	T_i	…	T_k	
y_{11}	y_{21}	…	y_{i1}	…	y_{k1}	
y_{12}	y_{22}	…	y_{i2}	…	y_{k2}	
⋮	⋮	…	⋮	…	⋮	
y_{1j}	y_{2j}	…	y_{ij}	…	y_{kj}	
⋮	⋮	…	⋮	…	⋮	
y_{1n}	y_{2n}	…	y_{in}	…	y_{kn}	
n_1	n_2	…	n_i	…	n_k	N
y_1	y_2	…	y_i	…	y_k	Y
$\bar{y}_1$	$\bar{y}_2$	…	$\bar{y}_i$	…	$\bar{y}_k$	$\bar{Y}$

THE MODEL

Analysis of variance

Fundamental concept of ANOVA is that **every value observed can be expressed** as the following function:

$$y_{ij} = \mu + \tau_i + e_{ij}$$

μ GRAND MEAN

scale common parameter to all the treatments, global mean

τ_i EFFECT OF TREATMENT I

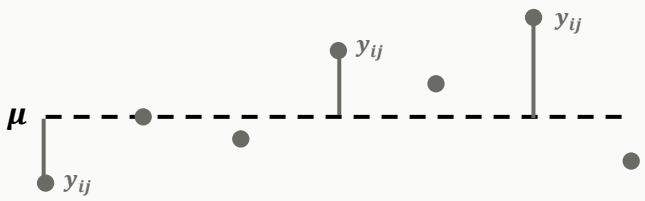
parameter that measures the effect in the treatment i

e_{ij} RANDOM ERROR

error assigned to each y_ij

Ronald Fisher asked a simple question: of all the variation we see in the data, how much is due to real differences between groups, and how much is just noise within each group?

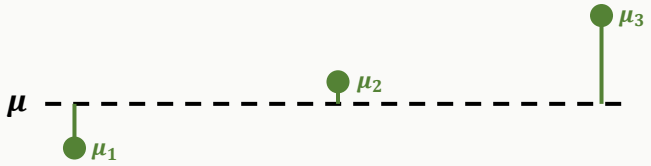
TOTAL VARIATION



$$y_{ij} - \mu$$

Each observation relative to the global mean

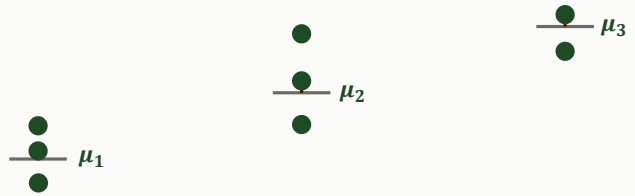
BETWEEN TREATMENTS



$$\tau_i = \mu_i - \mu$$

Each group's mean relative to the global mean

WITHIN TREATMENT



$$e_{ij} = y_{ij} - \mu_i$$

Each observation relative to ITS OWN group mean

Analysis of variance

Every value observed can be expressed

$$y_{ij} = \mu + \tau_i + e_{ij}$$

$$y_{ij} - \mu = \tau_i + e_{ij}$$

$$\tau_i = \mu_i - \mu$$

$$e_{ij} = y_{ij} - \mu_i$$

$$y_{ij} - \mu = (\mu_i - \mu) + (y_{ij} - \mu_i)$$

$y_{ij} - \mu$	**Total deviation** *(observation – global mean)*	$y_{ij} - \bar{Y}$
$\mu_i - \mu$	**Deviation of each treatment** *(treatment mean – global mean)*	$\bar{y}_i - \bar{Y}$
$y_{ij} - \mu_i$	**Deviation in the treatment** *(treatment observation – treatment mean)*	$y_{ij} - \bar{y}_i$

$$y_{ij} - \mu = (\mu_i - \mu) + (y_{ij} - \mu_i)$$ **Population**

$$y_{ij} - \overline{Y} = (\overline{y}_i - \overline{Y}) + (y_{ij} - \overline{y}_i)$$ **Sample**

IN TERMS OF THE VARIANCE

$$(y_{ij} - \overline{Y})^2 = [(\overline{y}_i - \overline{Y}) + (y_{ij} - \overline{y}_i)]^2$$

$$(y_{ij} - \overline{Y})^2 = (\overline{y}_i - \overline{Y})^2 + 2(\overline{y}_i - \overline{Y})(y_{ij} - \overline{y}_i) + (y_{ij} - \overline{y}_i)^2$$

$$\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \overline{Y})^2 = \sum_{i=1}^{k}\sum_{j=1}^{n}(\overline{y}_i - \overline{Y})^2 + \sum_{i=1}^{k}\sum_{j=1}^{n}2(\overline{y}_i - \overline{Y})(y_{ij} - \overline{y}_i) + \sum_{i=1}^{k}\sum_{j=1}^{j}(y_{ij} - \overline{y}_i)^2$$

0

$$\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \overline{Y})^2 = \sum_{i=1}^{k}\sum_{j=1}^{n}(\overline{y}_i - \overline{Y})^2 + \sum_{i=1}^{k}\sum_{i=1}^{j}(y_{ij} - \overline{y}_i)^2$$

$$\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij}-\overline{Y})^2 = \sum_{i=1}^{k}\sum_{j=1}^{n}(\overline{y}_i-\overline{Y})^2 + \sum_{i=1}^{k}\sum_{i=1}^{j}(y_{ij}-\overline{y}_i)^2$$

$$\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij}-\overline{Y})^2 = \sum_{i=1}^{k}\frac{(y_i)^2}{n}-\frac{Y^2}{N} + \sum_{i=1}^{k}\sum_{i=1}^{j}(y_{ij})^2-\sum_{i=1}^{k}\frac{(y_i)^2}{n}$$

Total Sum of Squares | **Sum of Squares between treatments** | **Sum of Squares in the treatments**

$$S^2_{Total} = S^2_{treatment} + S^2_{error}$$

VARIANCE ESTIMATOR

Between treatments

$$\frac{n\sum_{i=1}^{k}(\overline{y}_i-\overline{Y})^2}{K-1}$$

Within the treatments

$$\frac{\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij}-\overline{y}_i)^2}{N-K}$$

ANOVA Hypothesis

$H_0 : \mu_1 = \mu_2 = \ldots = \mu_k$ ⟶ All the treatment mean are equal

$H_1 : \mu_i \neq \mu_j$ ⟶ At least one treatment mean is different

Test statistic

$$\frac{S^2_{treatment}}{S^2_{error}} = \frac{\frac{n \sum_{i=1}^{k} (\bar{y}_i - \bar{Y})^2}{K - 1}}{\frac{\sum_{i=1}^{k} \sum_{i=1}^{j} (y_{ij} - \bar{y}_i)^2}{N - K}}$$

$K - 1$ ⟶ Degrees of freedom treatment

$N - K$ ⟶ Degrees of freedom error

$$\frac{S^2_{treatment}}{S^2_{error}} > F_{1-\alpha,\, df\ treatment,\, df\ error}$$ } Reject H_0

PUTTING IT TOGETHER

The ANOVA table

Everything we've seen is summarized in a single table:

Variation	Degrees of freedom	Sum of squares	Variance	F
Between treatments	$k-1$	$\sum_{i=1}^{k}\frac{(y_i)^2}{n}-\frac{Y^2}{N}$	$\frac{n\sum_{i=1}^{k}(\bar{y}_i-\bar{Y})^2}{K-1}$	$\frac{S^2\text{trat}}{S^2\text{error}}$
Within treatments	$N-k$	$\sum_{i=1}^{k}\sum_{i=1}^{j}(y_{ij})^2-\sum_{i=1}^{k}\frac{(y_i)^2}{n}$	$\frac{\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij}-\bar{y}_i)^2}{N-K}$	
Total	$N-1$	$\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij}-\bar{Y})^2$		